4 Recurrent neural networks

- so far, we encountered two types of data: tabular data and image data
- for the latter, we designed specialized layers to take advantage of the regularity in them
- in other words, if we were to permute the pixels in an image, it would be much more difficult to reason about its content
- most importantly, so far, we assumed that our data are all drawn from some distribution, and all the examples are independently and identically distributed (i.i.d.)

4 Recurrent neural networks

- unfortunately, this is not true for most data
- for instance, the words in this sentence are written in sequence, and it would be quite difficult to decipher its meaning if they were permuted randomly
- likewise, image frames in a video, the audio signal in a conversation, and the browsing behavior on a website, all follow a *sequential* order
- it is thus reasonable to assume that specialized models for such data will do better at describing them

4 Recurrent neural networks

- another issue arises from the fact that we might not only receive a sequence as an input, but rather might be expected to continue the sequence
- for instance, the task could be to continue the series $2, 4, 6, 8, 10, \ldots$
- this is quite common in time series analysis, to predict the stock market, the fever curve of a patient, or the acceleration needed for a race car
- again, we want to have models that can handle such data
- in short, while CNNs can efficiently process spatial information, *recurrent neural networks* (*RNNs*) are designed to better handle *sequential* information
- RNNs introduce state variables to store past information, together with the current inputs, to determine the current outputs